

The Blind Spot Paradox

The Blind Spot Paradox

When a self-repairing model destroys the evidence its own monitor needs

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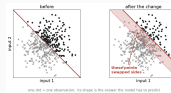
The Blind Spot Paradox: when a self-repairing model destroys the evidence its own monitor needs.

The Blind Spot Paradox

└ Concept drift: the rule changes, the data does not

First, the problem. A model learns a rule from data. Then the world changes, and the rule becomes wrong: that is concept drift. Here is the simple case we simulate. Points are split by a line. At the change, the line moves. The points in the shaded band switch labels, so a model trained before the change now gets them wrong.

Concept drift: the rule changes, the data does not



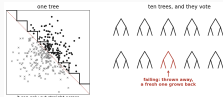
A model trained on the left keeps predicting, confidently, and is now **wrong** on the shaded band.

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└ Defence one: a model that repairs itself

The first defence is a model that repairs itself: the Adaptive Random Forest. It is made of ten decision trees that vote. A single tree can only draw step-shaped borders, so it is never perfect. When one tree starts making too many mistakes, it is thrown away and a new one grows in its place.

Defence one: a model that repairs itself



This is the **Adaptive Random Forest**. Ten trees, replaced one at a time.

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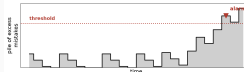
└ Defence two: a watchdog that raises the alarm

The second defence is a watchdog. Each mistake adds to a pile, each correct answer removes a little, never below zero. If the pile crosses a threshold, the alarm goes off and a human steps in. That alarm is all the human ever sees. So what happens when both defences run together?

Defence two: a watchdog that raises the alarm

The model answers, see observation at a time. (x = wrong)

x o o x o o o o x o o x o o x o o x o x o x



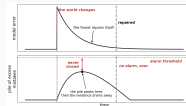
Two communities, two tools. What happens when both run at once?

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└ Put them together, and they cancel out

Here is what happens. The line moves, the mistakes start, and the pile begins to grow. But the forest replaces its trees, the mistakes stop, and the pile drains back to zero before it reaches the threshold. It is a race between the repair and the alarm. With a strict threshold, the repair almost always wins, and nobody is warned.

Put them together, and they cancel out



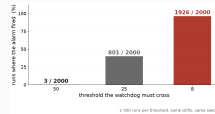
The repair has a date. The alarm never gets one. (schematic)

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└ Why not just lower the threshold?

So why not simply lower the threshold? We re-ran the paper's experiment at three thresholds, two thousand runs each. With the strictest one, the alarm went off three times in two thousand. With a lower one, it goes off almost every time. But a low threshold also goes off when nothing has changed, and an alarm that cries wolf ends up ignored. The paper concludes that no threshold is both trustworthy and sensitive.

Why not just lower the threshold?



A low threshold also fires when **nothing has happened**, and an alarm that cries wolf gets switched off.

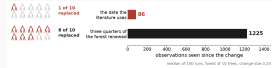
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└ Our contribution: when is the forest actually repaired?

To say who wins the race, you have to time the repair. The paper times it at the moment the first of the ten trees is replaced. But one tree out of ten is only the start of the repair. Wait until three quarters of the forest is renewed, and you wait fourteen times longer. The paper's stopwatch stops too early, and always in the forest's favour.

Our contribution: when is the forest actually repaired?

To call the race, you have to **time the repair**.



It fires at the first tree. It times the start of a repair, not its end.

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└ Measuring the repair in mistakes, not in trees

So we also measure the repair in mistakes. Same change as before. The first tree goes after 86 observations. The error is back to normal only after 261, and three quarters of the forest after 1,225. The change costs about 38 extra mistakes. And the paper's stopwatch says nothing about that cost, at any of the eighteen change sizes we tested.

Measure	What it counts	Value
First tree replaced t_{first}	the date the paper uses	86
Error back to normal t_{error}	the error settles back to its usual level	261
Forest 3/4 renewed t_{forest}	8 trees out of 10 replaced	1225
Excess mistakes $n(i)$	the extra errors the change caused	38
Watchdog's peak $t_{\text{watchdog}}(i)$	the highest point of the watchdog's pile	33

Same change as the previous slide, median of 100 runs. First three: observations after the change. Excess mistakes: counted over the next 2000 observations. Watchdog thresholds: 0, 25 and 50.

The paper's date says nothing about the damage.

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└─Where we stand

Where we stand. Yes: with the paper's strict threshold, the alarm really stays silent, and we measured how often. But the paper's stopwatch stops at the first tree, so the forest looks faster than it is, and the paper's figures are too optimistic. A better stopwatch is still an open question: we are testing candidates, and none has passed yet. Thank you.

Where we stand

at the paper's strict threshold

3 / 2 000

alarms

The blind spot is real.

the paper's stopwatch stops

up to 14×

too early

The repair looks faster
than it is.

a better stopwatch

?

candidates under test

The open question.

Thank you.